Krishna Patil

ee24btech11036@iith.ac.in | krishnapatil12.k4@gmail.com | +91 9764003122 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.96
Vishwanathappa hurne junior College	Class XII (Maharashtra State Board)	2024	83.50%
Oxford International School,Nanded	Class X (CBSE board)	2022	95.40%

TECHNICAL SKILLS

Languages & Scripting: : Verilog | SystemVerilog (RTL) | C | Python | Bash | Zsh

EDA Tools: Vivado | LibreLane | OpenROAD | Cadence Virtuoso | Calibre | XSchem | Magic | NGspice | netgen | LTspice

PCB Design & CAD: KiCad | EasyEDA | Autodesk Fusion

PROJECTS

NeuraCore: Custom CNN Inference Accelerator Development Board - *SystemVerilog, Zynq-7000 SoC*

- Led a 4-person engineering team to architect an **Eyeriss-inspired**, **row-stationary systolic accelerator** on a Zynq-7000 SoC
- Engineered an **8x8 NoC** with **hierarchical 2x2 clustering**, scaled down from a 3x60 target due to **400 LUTs/PE** cost on the FPGA
- Verified the **NoC** via **SystemVerilog testbenches** and reg-based debug, achieving **timing closure** under **synthesis constraints**

RISC-V Processor Chain: Pipelined RV64IM and RV64I - *Verilog, Artix-7*

- Designed a series of **RISCV** cores with **data forwarding** , **branch prediction** and **load-use hazard detection** for the **M-extension**
- Raised **pipelined RV64IM** to **35 MHz stable** by tracing **TNS/WNS violations** and replacing async logic with sync equivalents
- Verified via **Verilog testbenches** on Artix-7-100T **(14K LUTs, 32 BRAMs, 54 DSPs**) across stress test programs

Adder PPA Characterization: RCA, CLA, Brent-Kung, Kogge-Stone - *LibreLane, Sky130*

- Hardened four **32-bit adder architectures** through the **RTL-to-GDSII flow**, clearing **DRC** and **LVS** on all synthesis runs
- Closed timing across all four architectures via **binary-search clock period sweeps**, hardening each design through RTL to GDSII
- Found **RCA area exceeds CLA under tight timing**, as the tool inserts **buffering/upsized cells** to meet **aggressive clock targets**
- Compared area/delay-optimized synthesis across all architectures; Kogge-Stone hit **472 MHz** vs RCA **229 MHz** at **2.5×** the area

SAR ADC & R-2R DAC - *Breadboard, Arduino*

- Built a 5-bit ***SAR* ADC** (Arduino-based successive-approximation logic, R-2R DAC, comparator) and a standalone **R-2R ladder DAC** on breadboard, handling 0-5V analog inputs
- Verified functional correctness by digitizing and **reconstructing sine** and square wave inputs, confirming conversion tracked the input waveform shape across the input range

Mini Projects

- Constructed **op-amp integrator/differentiator** circuits and **common-source/common-emitter** discrete transistor amplifiers.
- Built a **555 timer IC** from the transistor level up, differential amp to comparator to full 555 topology, simulated in

LTSpice.
- Designed a **multi-mode** digital clock with alarm, stopwatch, and 12/24-hour conversion.
- Built a traffic light **FSM controller** for a T-junction intersection.
- Wrote a **behavioral convolution** implementation using **for-loop constructs**, synthesized directly by the tool.

PUBLICATIONS / RESEARCH

Scientific Computing on Hardware, *G.V.V. Sharma, Krishna Patil, Subhodeep Chakraborty, Nara Prajwal*

IEEE VLSI-SATA 2026 (Accepted)

- Implemented an RK4-based ODE solver and expression parser on an ATmega328P, fitting the app in **12.3 KB** Flash.
- Achieved **2.38×10^{11}** ULP max error vs. an MPFR reference, orders of magnitude below Euler-method baselines.

CMOS Image Sensor Research & Development Lab, *Prof. Kapil Jainwal, IIT Hyderabad*

- Redesigning **correlated double sampling** and **ramp generator blocks (180nm)** for a fabricated image sensor tape-out
- Studied CMOS image sensor architectures (**3T/4T APS**), C**DS theory**, and **pixel readout** techniques
- Practiced **full schematic-to-layout flow** in **Cadence Virtuoso**, characterizing an inverter via **NMOS/PMOS W/L tuning**

POSITIONS OF RESPONSIBILITY

Internship Coordinator | Office of Career Services (OCS), IIT Hyderabad

- Coordinated recruiter outreach and interview logistics for the internship drive across hardware and core engineering companies.

Head TA, Matrix Theory (EE1030), IIT Hyderabad

- Designed a course project on image compression via **truncated SVD**, requiring students to reconstruct low-rank approximations and analyze the k vs. reconstruction-error trade-off using the **Frobenius norm**.

Core Member | Electronica - Electronics and Signal Processing Club

- Led a team in the eYantra Robotics Hackathon, building a maze-solving robot that advanced to Stage 2; ran technical sessions and junior orientation for the club.

RELEVANT COURSEWORK

Computer Architecture | Digital Systems | Introduction to VLSI Design | RTL Design & Verification | Analog Electronics | Electronic System Design Project Lab | Introduction to CMOS Processing | Analog IC Design | Matrix Theory | Signal Processing | Vector Calculus